The Fundamental Theorem of Algebra says many very important things about math. The most important thing for Math 3 that it says is that any polynomial of a degree *n* also must have *n* roots. If this doesn't make sense, look at this example:

$2x^2 + 3x - 5$ can be factored into $(2x + 5)(x - 1)$, meaning its **zeros**, or **roots** (where the function hits the x-axis) are where $2x + 5 = 0$ and $x - 1 = 0$, which is when $x = -5/2$ and $x = 1$.
There are 2 roots, which is the same as the equation's degree. Its degree is 2 because its largest exponent is 2.

Some practice:

$3x^3 + 2x^2 - 5$:
Degree: 3
Number of roots: 3
$5x^2 + 3x^5 + 3 + 2x$:
Degree: 5 (even though the terms aren't placed descending order based on their exponent)
Number of roots: 5

Now for the important part: a polynomial of degree *n* has *n* roots, but some (or all) of these roots could be **complex roots**, which are roots that equal a **complex number**. You should have covered complex numbers before, but we'll briefly go over them. They are numbers that contain a real number and an imaginary number (i, or $\sqrt{-1}$)
A complex root will be in the form $x = a + bi$. An example is $x = 3 - 2i$.

Here is an example of an equation with complex roots: $x^2 + 9 = 0$

At first glance, this seems unsolvable. But if we use imaginary numbers, we can get a solution:

$x^2 = -9$
$x = \pm 3i$, which can also be written as $x = 3i$ and $x = -3i$. (There are 2 roots, as we can see.)

The factored form of the equation is $(x + 3i)(x - 3i)$.

The next important thing you must know is that complex roots always come in pairs. If we look at the above example, the roots are the same, except the addition sign is changed to a subtraction sign. These are called **conjugate roots**.

So if a root in the general form a + bi shows up, there must also be a root a - bi.

When conjugate pairs multiply together, the imaginary parts cancel out because it is a difference of 2 perfect squares.

$(a + bi)(a - bi)$

$a^2 - abi + abi - b^2i^2$

$a^2 - b^2i^2 = a^2 + b^2$

This is the reason why if you have an equation with one imaginary root, it must also have the root's conjugate, because otherwise the equation would have imaginary numbers in it. (You don't have to have conjugate roots in an equation that does have imaginary numbers, but in this class you'll only deal with equations with real numbers.)

Dividing Polynomials

There are multiple ways to divide polynomials, the easiest way is to reduce complex equations to the **factored form**, and eliminate common factors that appear in both the numerator and the denominator.
However, not all equations can be factored easily, and in many cases, equations have remainders. To solve for these types of equations, we have two key formulas.

I. **Polynomial Long Division**

This method functions just like regular long division, and any polynomial can be divided by another polynomial of the **same or lower degree**. Meaning that you can divide:

- quadratic equations / linear, or quadratic equations
- cubic equations / linear, quadratic, or cubic equations

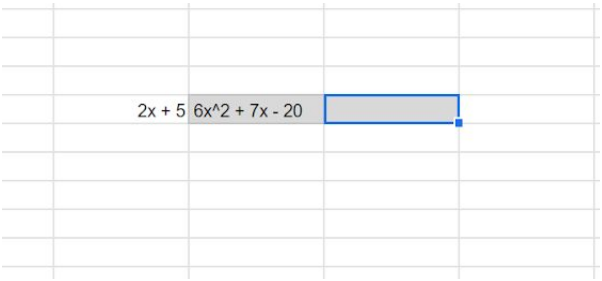
And so on...

Solving for Long Division

Let's practice with the problem:

$$6x^2 + 7x - 20 / 2x + 5$$

1) Set up your division statement, numerator inside the box, and denominator outside the box.

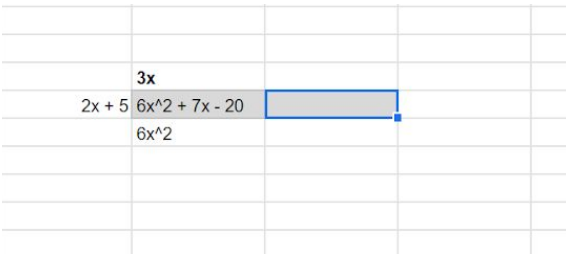


2) Start with the first term of both the numerator and the denominator, in this case, 2x and $6x^2$. Ask:

What multiplied to 2x will equal to $6x^2$?

- It would be 3x: (2x)(3x) = $6x^2$

Just like long division, write 3x on top of the box, and $6x^2$ under the $6x^2$ inside the box, as they both share the variable degree x^2.



3) Since we are dividing by a binomial, whatever factor is applied to the first value, **must** be applied to the second. So: multiply (3x)(5) =

- It would be 15x

So write 15x under 7x, because they both share the same variable degree: x^1

	3x	
2x + 5	6x^2 + 7x - 20	
	6x^2 + 15x	

4) Now just like in long division, we have to subtract the result of our new factor from the numerator. Since we multiplied positive integers together, our result was also positive. So we know that in order to cancel these values out, we need to **SUBTRACT.**

- $6x^2 - 6x^2 = 0$, so we can cancel those out.
- 7x - 15x = -8x, which becomes our new simplified numerator.

	3x	
2x + 5	6x^2 + 7x - 20	
-	(6x^2 + 15x)	
	-8x	

5) Now that we have canceled out one value, we can bring down another, bring -20 down on the same level as -8x, just like in regular long division.

	3x	
2x + 5	6x^2 + 7x - 20	
-	(6x^2 + 15x)	
	-8x - 20	

6) Repeat the same process: Ask, **What multiplied to 2x is equal to -8x?**

- -4: (2x)(-4) = -8x

Multiply both parts of the binomial with this new value:

- (5)(-4) = -20

	3x - 4	
2x + 5	6x^2 + 7x - 20	
-	*(6x^2 + 15x)*	
	-8x - 20	
	-8x - 20	

7) Subtract the Results

- (8x + 20) - (8x + 20) = 0

This division is complete! Since all values were canceled out and all degrees of x were accounted for.

	3x - 4	
2x + 5	6x^2 + 7x - 20	
-	*(6x^2 + 15x)*	
	-8x - 20	
-	*(-8x - 20)*	
	0	

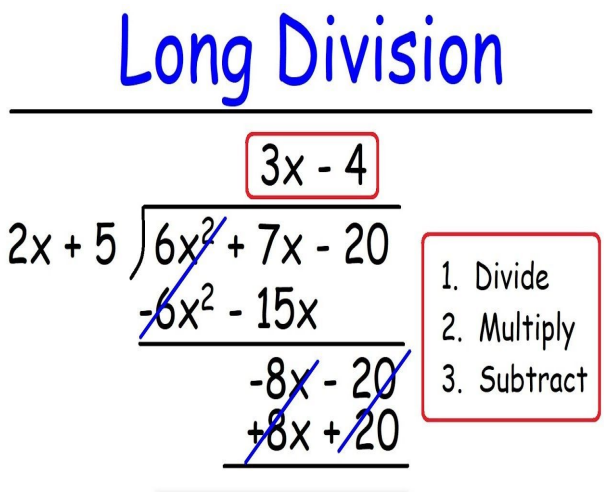
8) Double check the results

Always be sure to double check your work. We can make sure our solution is correct by multiplying, since it's the opposite of dividing, our denominator by our solution.

- FOIL (3x-4)(2x+5)
- The answer is $6x^2 + 7x - 20$

Which means that our answer was correct!

Here's a summary of what we did:



Now, dividing polynomials will not always be so clean, often, after canceling out all but one value, you'll be left with a remainder. You know you've hit a remainder when you **can't** answer the question:

What multiplied to <u>the first value of my denominator</u> will equal <u>this?</u>

When you realize you cannot divide any further, your answer is simply whatever solutions you have already determined, like 3x - 4, plus/minus the remainder value/denominator

For example, had there been a remainder of 1 in this problem:

The solution would have looked like - $(3x - 4 + \frac{1}{2x+5})$

II. The Remainder Theorem

This is where the Remainder Theorem comes into play, often, you might want to know just the remainder of a problem, or if a remainder **exists at all**, which would be a remainder of 0.

For this, we use the Remainder Theorem.

All the Remainder Theorem states is that: the **solution** to the denominator **substituted** into the numerator will **yield** the remainder.

Or simply: **Plug in x for your numerator into your denominator.**

Let's use the same problem: $6x^2 + 7x - 20 / 2x + 5$

1) Solve for the denominator: what is x?

$$2x + 5 = 0$$

$$2x = -5$$

$$x = -5/2$$

2) Now we can substitute this solution (-5/2), into our numerator:

$$6(\tfrac{-5}{2})^2 + 7(\tfrac{-5}{2}) - 20$$

3) Subtract:

$$37.5 - 17.5 - 20 = 0$$

4) This theorem is complete!

We know that this equation has no remainder, which is why the remainder theorem yielded 0.

Practice Problems:

First Solve for the Remainder, and then Divide using Polynomial Long Division.

https://www.mesacc.edu/~scotz47781/mat120/notes/divide_poly/long_division/long_division_practice.html